



Deconstruct artificial intelligence's productivity impact: A new technological insight

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ABSTRACT

Some viewpoints suggest that IT investment seems to fail to significantly stimulate enterprise productivity in some cases. Therefore, revealing the impact of AI on firm productivity is an important topic to analyze whether Solow's paradox can be valid in the digital age. Based on panel data of 3235 listed companies in China from 2007 to 2021, we comprehensively discuss the impact and mechanism of AI on firm productivity using fixed-effects model, systematic GMM model, and mediated-effects model. Key findings include: AI significantly improves firm productivity, especially in state-controlled, internationally minded, and innovative firms. Mitigating information asymmetry is a key channel, while specialized division of labor and independent green innovation are potential ones. Supply chain digital transformation policies enhance the productivity effect of AI, and AI shows green development benefits. Additionally, the dynamic decomposition effect shows that the productivity-enhancing effect of AI is slowing down in the long run. This research provides important insights into understanding AI's role in the digital age and holds significance for firms and policymakers.

1. Introduction

In the late 1980s, American scholar Strassman surveyed 292 enterprises and found a strange phenomenon. There was no obvious correlation between the IT investment of these enterprises and the return on investment (ROI). Robert Solow, an economist who won the Nobel Prize in 1987, called this phenomenon the "productivity paradox": "Computers everywhere except in the productivity statistics." The central argument is that despite the enormous advances made by firms in the field of science and technology, technological progress has not been translated into corresponding productivity gains throughout the economic system [1]. According to Solow, there are three broad reasons for this phenomenon. One is the delayed effect, which means that technological innovations usually take some time to fully affect productivity. The second is the problem of resource allocation. If there are problems in resource allocation by firms or Governments, this may affect the application and diffusion of new technologies, thus limiting productivity gains [2]. Thirdly, there is the issue of measurement. In emerging industries and high-technology areas, traditional productivity measurement methods may be difficult and inaccurate, resulting in a failure to scientifically reflect the benefits of technological progress. The

introduction of Solow's paradox has led to extensive discussion and research in economics, prompting greater attention to the actual impact of technological innovation on productivity and economic development. For example, green technology innovation and renewable energy technology innovation [3,4].

In fact, Solow was based on a sample of U.S. firms from 1980 to 1990, and the study found that the information and communication technology industry (ICT) seemed unable to promote productivity growth. However, in the 21st century, scholars have demonstrated that Solow's paradox can be overcome. For example, Madan and Ashok [5] state that the use of AI in government represents a fundamental shift in governance, contributing to lean government and the delivery of efficient, personalized services, and is at the forefront of governance in the digital age (DEG), even though it can be risky and ethically challenging. Building on this foundation, Madan and Ashok [6] further explored issues related to the adoption and proliferation of AI in public administration through a systematic literature review. The study points out that the implementation of AI involves technological, organizational, and environmental factors and that there are tensions between automation and augmentation, facilitation and autonomy, and data accessibility and security and privacy. Institute [7] assessed the scale and automatable

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